USB_CAN TOOL Debug Software

Manual

Manual Version: V2.05

Update Date: 2023.07.01

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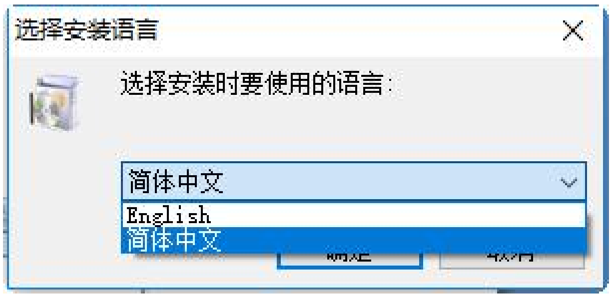
Chapter 1 Software Installation

1.1 USB-CAN TOOL Software Installation

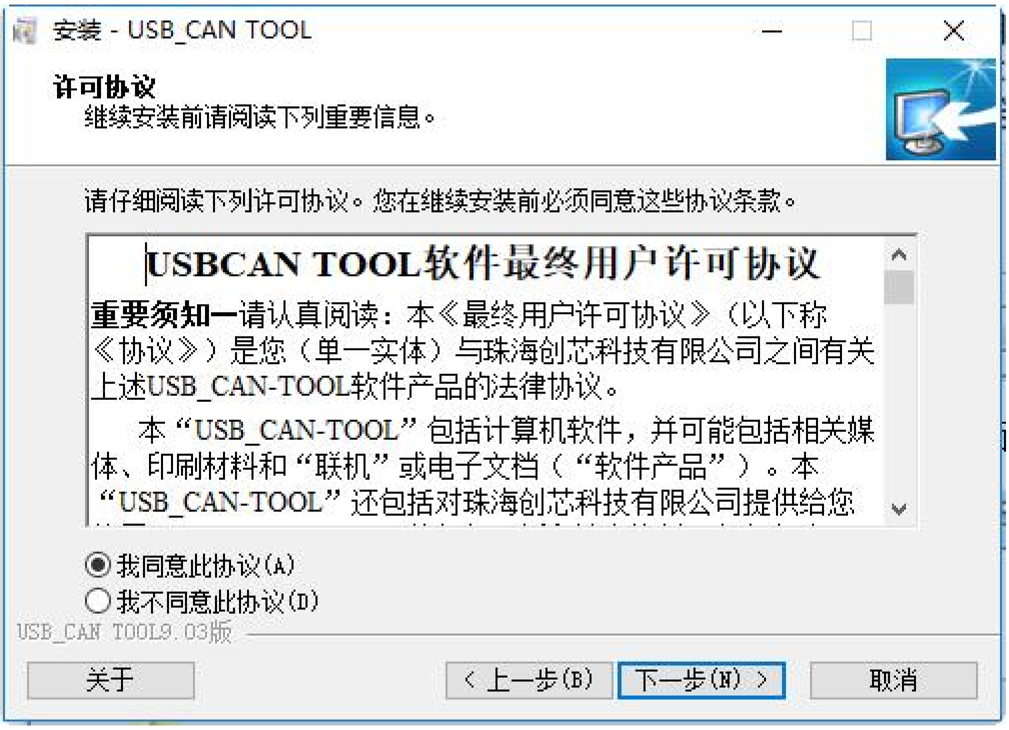
Directly run the "Packages\Debugging Tools\Original Debugging Tools\USB_CAN TOOLSetup(V9.xx).exe" in the directory of the package to install it.

Run the installation package and the screen shown below will appear.

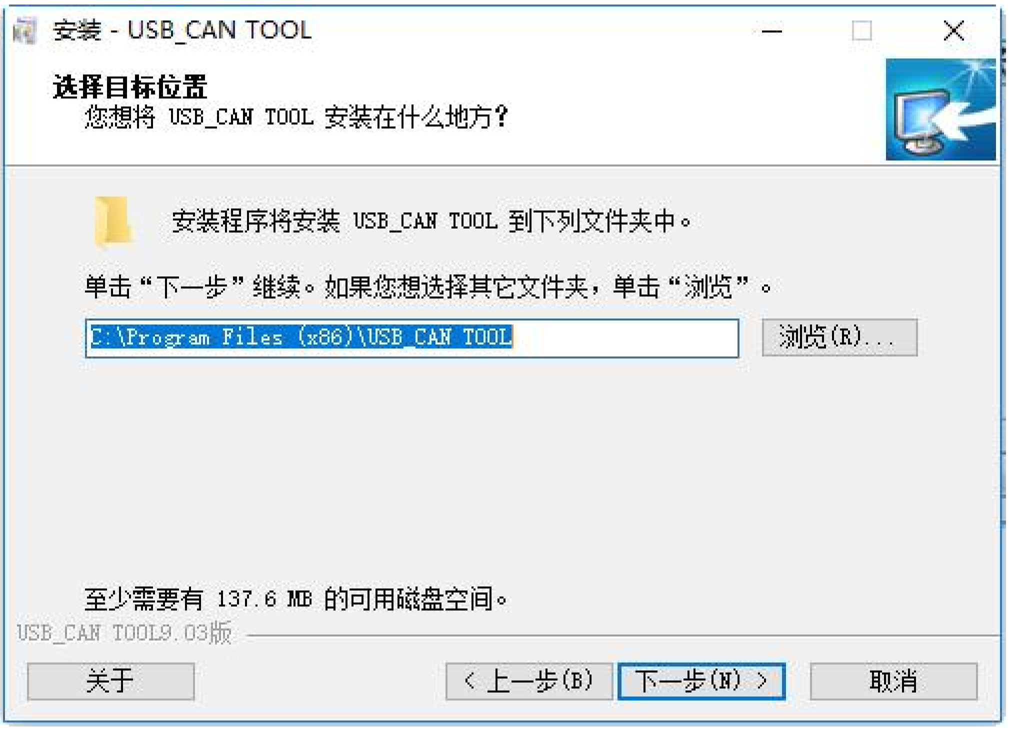


Select the corresponding language first.

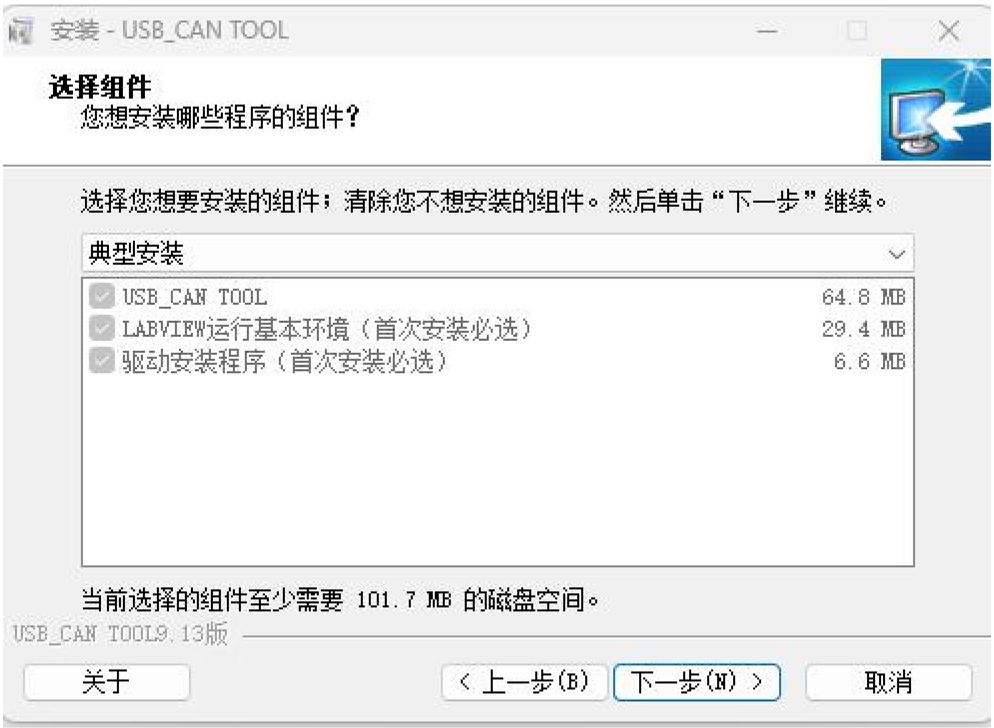
Click Next to enter the license agreement confirmation screen.



Check the "I agree to this agreement" option and click Next. Enter the installation path selection interface



Select the desired installation path and click Next to enter the installation component selection interface.



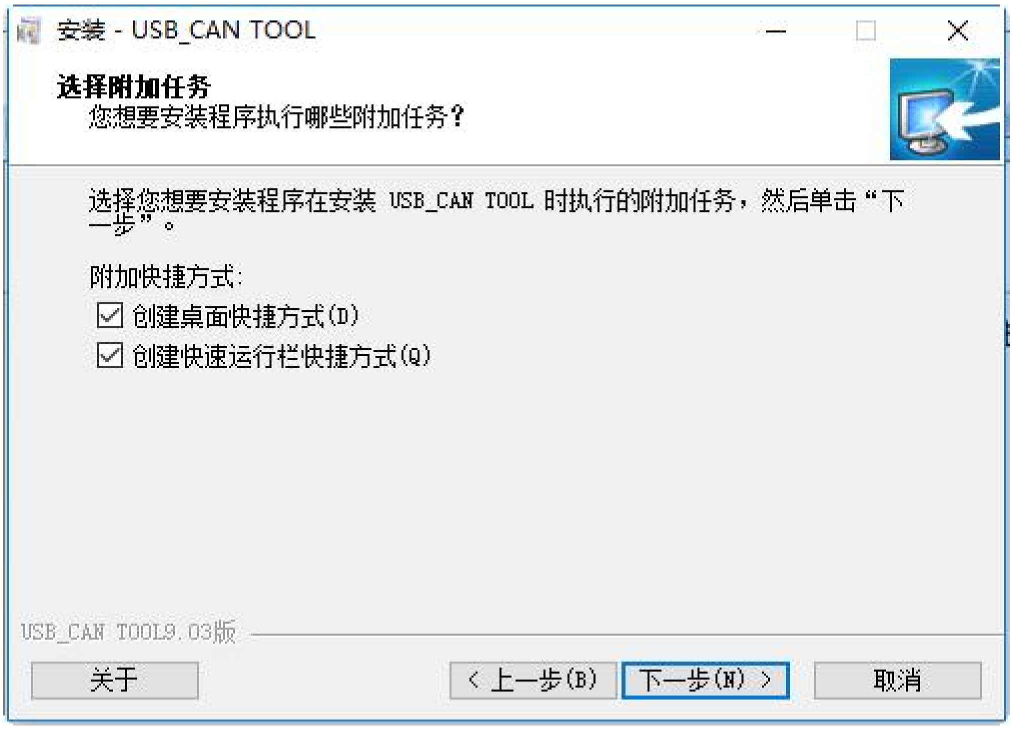
USB_CAN TOOL: debugging software, mandatory, checked by default.

LABVIEW Running Basic Environment: mandatory for the first installation, this is the running environment installation package required for applications developed on the LABVIEW platform.

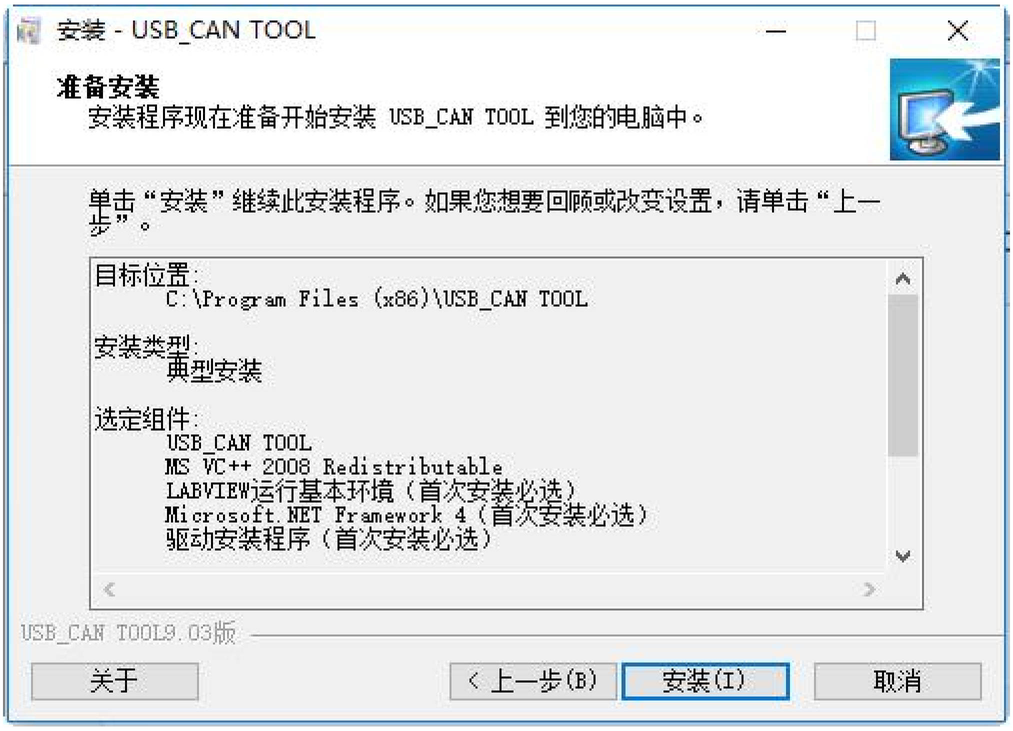
Driver Installer: USB driver.

Note: The system has checked all the boxes and will be installed automatically and silently.

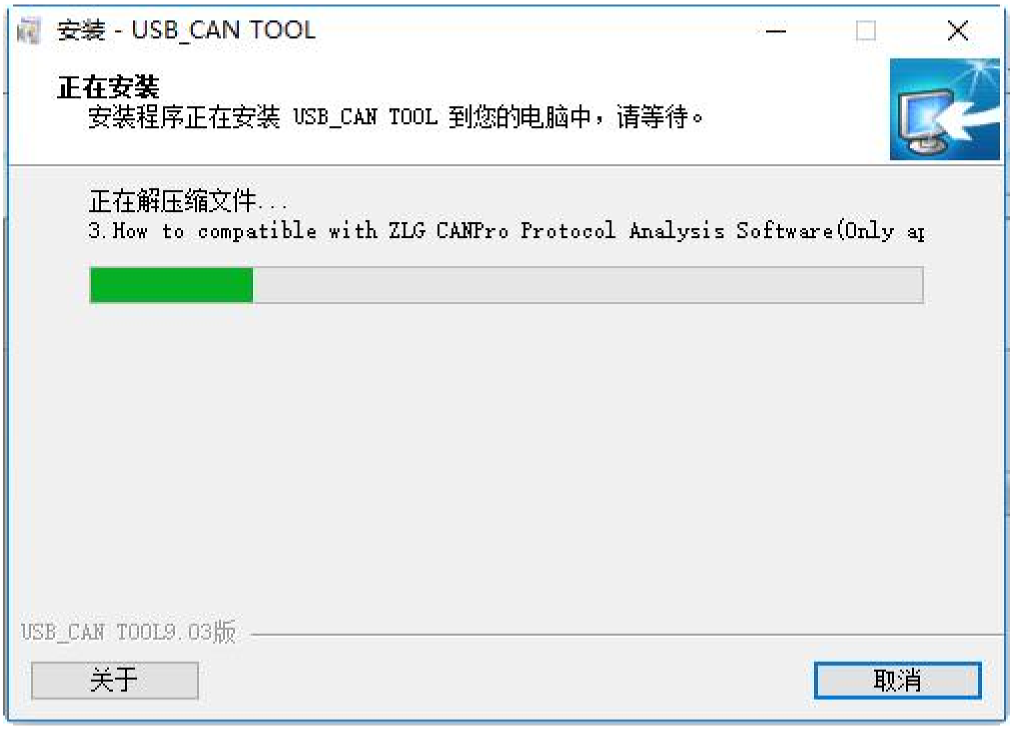
Click Next.



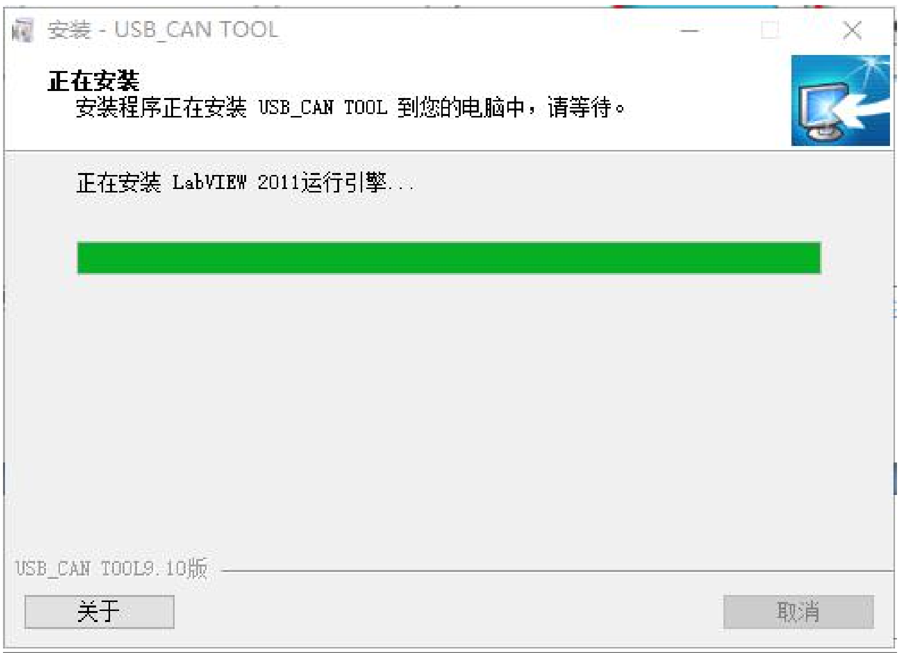
Select as needed and click Next.



Click on "Install". The first USB_CAN TOOL software installation begins.



The installer automatically enters the NI LabVIEW Run-Time Engine 2011 SP1 runtime environment installation.



Finally , click to install the driver and close the software.


安装 - USB_CAN TOOL
正在安装
安装程序正在安装 USB_CAN TOOL 到您的电脑中，请等待。
正在安装 USB驱动...
Device Driver Installation Wizard
Welcome to the Device Driver Installation Wizard!
This wizard helps you install the software drivers that some computers devices need in order to work.
To continue, click Next.
< 上一步(B)
下一步(N) >
取消

This concludes and completes the installation of the entire software.

Chapter 2 USB-CAN Tool User's Guide

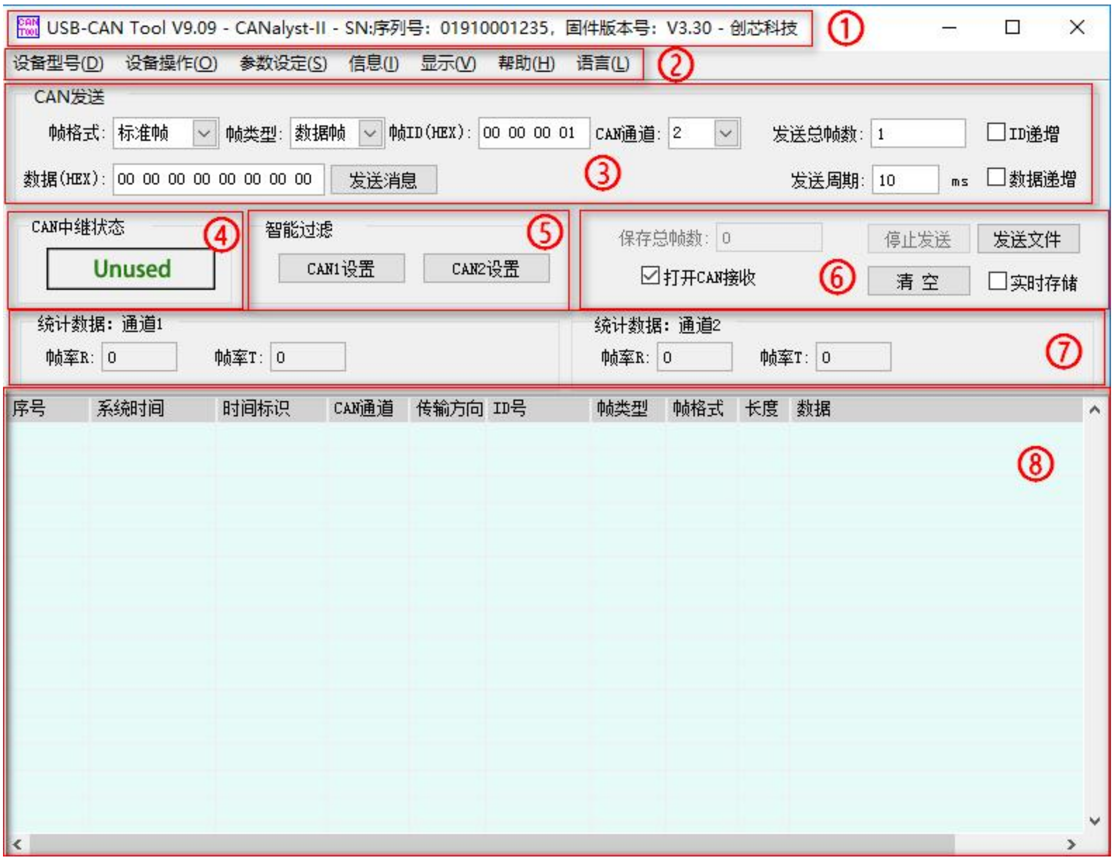
If terminating resistors are needed, USBCAN please short R+ and R- with wires, and CANalyst-II analyzer please turn the dip switch to ON position.

Note: Two 120-ohm terminating resistors must be guaranteed on the normal CAN bus, otherwise it will affect the normal operation of the CAN bus.

Connect the device to the USB port of the PC via the USB connection cable;

When you run USB-CAN Tool for the first time, because of its operation on the registry (this operation has LabVIEW runtime engine to complete automatically), it may lead to the interception of 360 firewall and other software, please select and confirm as shown in the figure below, otherwise it may cause unpredictable errors after the program is run.

2.1 Introduction to the interface



2.1.1 Title Bar

USB-CAN Tool V9.xx is displayed when the device is turned off, where V9.xx indicates the current program version number; when a USB-CAN adapter is successfully turned on, the model number, serial number, firmware version number, and brand name will be appended to the above name.

2.1.2 Menu Bar

Most of the functions of this tool are organized in the menu, and the functions are realized by clicking the corresponding menu.

2.1.3 Send Setting Area

This area contains setting information related to CAN message sending, among which the function of "ID increment" is that when sending multi-frame data, the ID value of the next frame is 1 larger than that of the previous frame; the function of "data increment" is that the 8 bytes of data sent are composed of 8 bytes in the order of low to high bytes. The function of "data increment" is to form a 64-bit number by sending 8 bytes of data in the order of bytes from the lowest to the highest, and automatically add 1 to the number every time the data is sent.

2.1.4 CAN relay status

The CAN relay function is configured via the menu "Device Operation" -> "Relay Mode Options". When configured in relay mode, the relay status is displayed here. If you want to disable the relay function, you need to disable it through the menu "Device Operation"->"Relay Mode Option".

Note: When the relay function is turned on, the baud rate and other parameters are invalid, but the transmit and receive and intelligent filtering functions can be used normally.

2.1.5 Intelligent Filtering

The two channels can be set separately, the filtering is only for receiving, after enabling filtering, the ID in the filter list or ID in the ID segment will be displayed by receiving, and the data outside the filter list will be blocked and discarded. With the flexible setting of intelligent filtering, it is possible to allow receiving or blocking to receive any ID or ID segment.

2.1.6 Control buttons

"Stop sending" - stops the current sending operation; "Send file" - stores the sent frame information in a Send file" - store the sent frame information in a specific format in a file and send it in the order of the frames stored in the file; "Turn on CAN receive" - turn on CAN receive if it is checked, otherwise the CAN receive function is in the "hang" state, at this time the host computer is in the "hang" state. Otherwise, the CAN reception function is in "hang" state, at this time, the upper computer no longer displays the CAN bus data received by the USBCAN device, but the USBCAN device is still receiving data on the CAN bus; "Clear" - clear the contents of current Clear" - clears the contents of the current data list; "Real-time storage" - saves the real-time data to a file.

Note: Sending a file can be done in the same format as the file saved by the real-time storage function, and the saved file can be sent out directly in the form of a file.

2.1.7 Statistics Area

Includes unidirectional rate and receive incremental checksum error counters for channel 1 and channel 2. "Frame rate R" - receive rate in fps (frames per second); "Frame rate T" - transmit rate in fps (frames per second).

2.1.8 Data List

A list of data transmission and reception information is displayed.

2.2 Starting Up and Shutting Down the Device

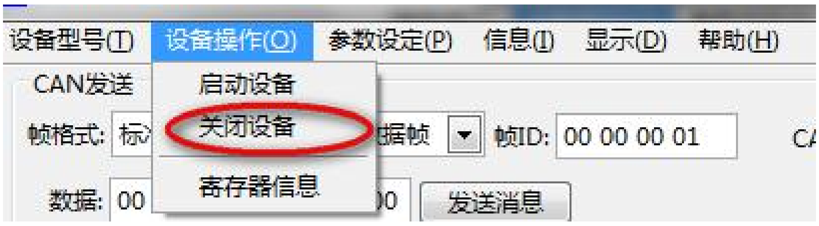
2.2.1 Selecting the Device Type

Select the corresponding device model in the "Device Model" drop-down menu in the menu bar, where: "USB-CAN" is a single-channel USBCAN adapter, "USB-CAN2.0" is a dual-channel USBCAN adapter. USB-CAN" is a single-channel USBCAN adapter and "USB-CAN2.0" is a dual-channel USBCAN adapter. USB-CAN2.0" is selected for all models of our products.

2.2.2 Starting the USBCAN Adapter

Select "Start Device" in the "Device Operation" drop-down menu in the menu bar and the program will automatically find and turn on the USB-CAN adapter.

2.2.3 Shutting Down the USBCAN Adapter



Turn off the open USB-CAN adapter by selecting "Turn Off Device" in the "Device Operation" drop-down menu on the menu bar.

2.3 Sending Messages

Set up the sending conditions in the sending setting area: frame format, frame type, frame ID, CAN channel, total number of frames sent, whether ID increment, whether data increment, sending cycle and data, and then press the Send Message button to start sending; here we emphasize on explaining the frame ID, total number of frames sent, ID increment, data increment and data.

2.3.1 Frame ID

That is, the ID value of CAN message frame, this tool adopts the right-aligned method, the ID value is inputted in 4 bytes, expressed in hexadecimal, and the bytes are separated by space, for example, if ID=0x18FF0023, the frame ID should be written as: "18 FF 00 23";

2.3.2 Send total number of frames

Set the total number of frames to be sent this time, value=-1 means no limitation on the number of frames to be sent (i.e., continuous sending), value=0 is invalid, value>=1 means the specific number of frames to be sent, and the sending of frames will stop automatically when the number of frames to be sent reaches the set value;

2.3.3 ID increment

When the total number of frames sent is greater than 1, if the ID increment item is checked, the value of the currently sent frame ID will be added 1 to the value of the previously sent frame ID, such as ID sequence: "00 00 00 01", "00 00 00 02", "00 00 00 03" ...... (right aligned)

2.3.4 Data increment

"10 00 00 00 00 00 00 00 00 00 00" ...... and so on; (display mode: low first, then high)

Note: The above data fields are all in hexadecimal representation.

In the CAN transmit data input box, from left to right, it is the 1st byte to the 8th byte of the data field in the CAN message, e.g.: fill in "01 23 45 67 89 AB CD EF" in the input box, and the 64-bit unsigned number from low to high is 0xEFCDAB8967452301. In the CAN message list, first low and then high, it is 0xEFCDAB8967452301. In the CAN message list, "01 23 45 67 89 AB CD EF" is displayed first (default display); "EF CD AB 89 67 45 23 01" is displayed first.

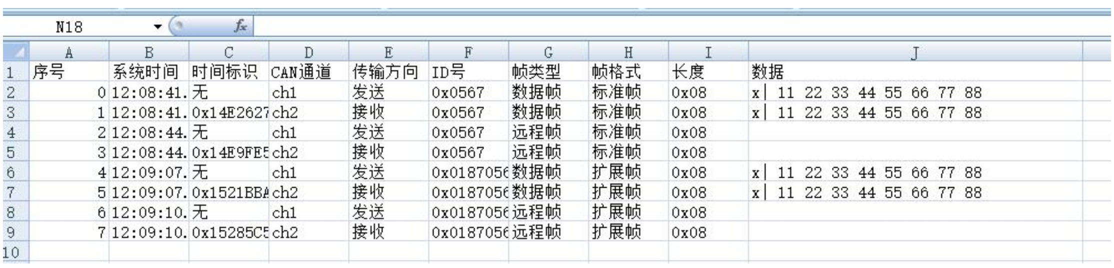
(The display mode is available under the display menu.)

2.3.5 Data

When filling in the data, the low byte is on the left and the high byte is on the right. The CAN message data field contains up to 8 bytes of data, so the standard format for filling in the data is 0 to 8 hexadecimal digits spaced by spaces (the maximum value of each digit is 0xFF), **and the length of the data field is judged automatically according to the number of bytes filled in**; if the option of "data increment" is checked, the length of the data field is locked to 8 bytes. If the "Data Increment" option is checked, the length of the data field is locked to 8, if the number of input bytes is not enough to 8, then the high byte direction will automatically fill in zero to 8 bytes, for example, if the data is filled in as "12 34 56 78", it will be automatically supplemented as: "12 34 56 78 00 00 00 00". (low first, then high)

2.3.6 Sending files

Click the "Send File" button, as shown below, to select and start sending the file.



	A	B	C	D	E	F	G	H	I	J
1	序号	系统时间	时间标识	CAN通道	传输方向	ID号	帧类型	帧格式	长度	数据
2	0	12:08:41.	无	ch1	发送	0x0567	数据帧	标准帧	0x08	x| 11 22 33 44 55 66 77 88
3	1	12:08:41.	0x14E2627	ch2	接收	0x0567	数据帧	标准帧	0x08	x| 11 22 33 44 55 66 77 88
4	2	12:08:44.	无	ch1	发送	0x0567	远程帧	标准帧	0x08	
5	3	12:08:44.	0x14E9FE5	ch2	接收	0x0567	远程帧	标准帧	0x08	
6	4	12:09:07.	无	ch1	发送	0x0187056	数据帧	扩展帧	0x08	x| 11 22 33 44 55 66 77 88
7	5	12:09:07.	0x1521BBA	ch2	接收	0x0187056	数据帧	扩展帧	0x08	x| 11 22 33 44 55 66 77 88
8	6	12:09:10.	无	ch1	发送	0x0187056	远程帧	扩展帧	0x08	
9	7	12:09:10.	0x15285C5	ch2	接收	0x0187056	远程帧	扩展帧	0x08	
10										

The format of the send file is a csv file, each line in the file is a frame of data containing 10 elements:

1、serial number: starting from 0.

2、system time: real-time storage of the saved file, the system time display of the PC clock, send the file, will be sent to the system time interval. When there is no time system time, it will be sent with the set value of the sending period of the interface.

3、Time mark: the real-time storage of the saved file, the time mark indicates the exact time of reception, the power-up time as a starting point, the unit 0.1ms. for sending, invalid.

4、CAN channel: real-time storage of the saved file, there is an indication of the CAN channel, for sending, invalid.

5、Transmission direction: real-time storage of saved files, there is an indication of the transmission direction, for sending, invalid.

6、ID：Beginning with 0x, expressed in hexadecimal.

7、Frame type: data frame/remote frame.

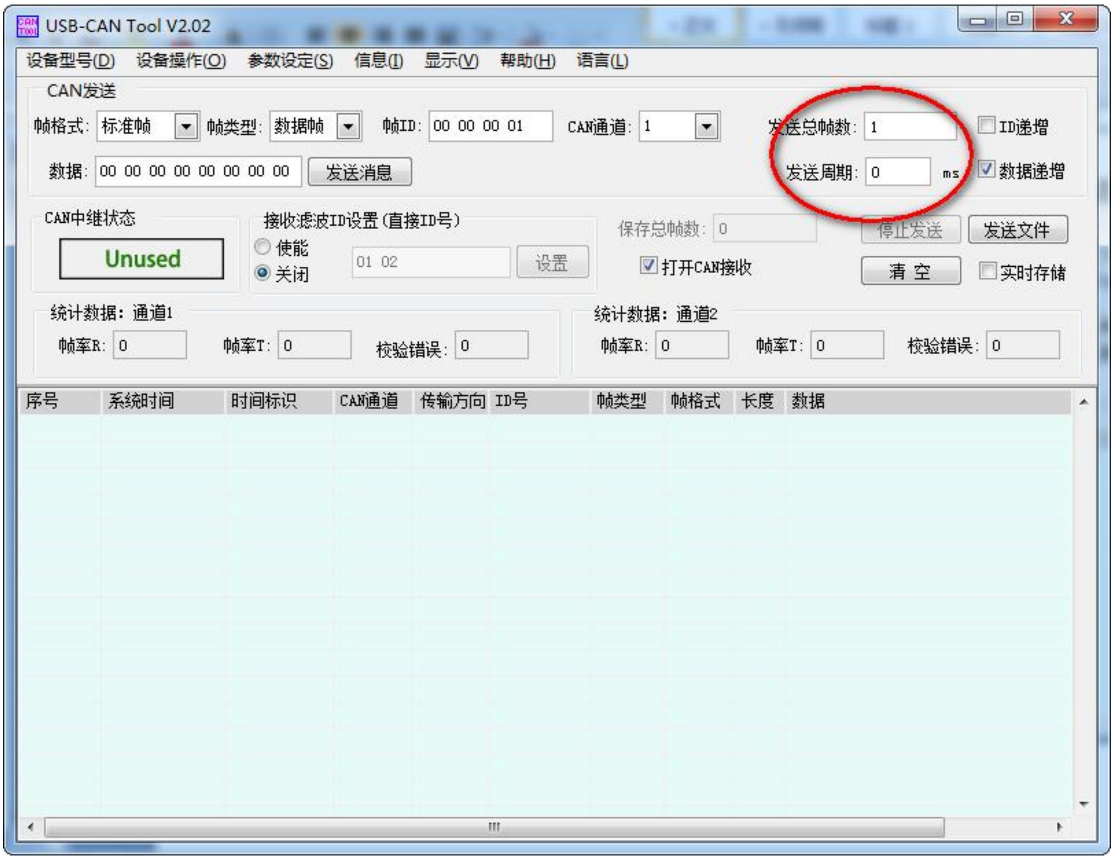
8、Frame format: standard frame/extended frame.

9、Length：Beginning with 0x, expressed in hexadecimal. 1-8 bytes.

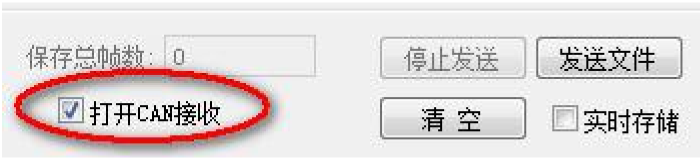
10、Data：x|beginning, expressed in hexadecimal. 1-8 bytes.

Note: For specific samples, please refer to the CD-ROM "CD-ROM\Example For Send File.csv"/or the software installation directory: "C:\Program Files (x86)\USB_CAN TOOL\" for reference. Example For Send File.csv" for reference. You can also store a file in real time and then modify the file.

6、When sending files, the sending period in the main interface of the software is valid and needs to be configured according to the needs.


USB-CAN Tool V2.02
设备型号(D) 设备操作(O) 参数设定(S) 信息(I) 显示(V) 帮助(H) 语言(L)
CAN发送
帧格式: 标准帧
帧类型: 数据帧
帧ID: 00 00 00 01
CAN通道: 1
发送总帧数: 1
ID递增
数据: 00 00 00 00 00 00 00 00
发送消息
发送周期: 0 ms
数据递增
CAN中继状态
Unused
接收滤波ID设置(直接ID号)
使能
关闭
01 02
设置
保存总帧数: 0
打开CAN接收
停止发送
发送文件
清 空
实时存储
统计数据: 通道1
帧率R: 0
帧率T: 0
校验错误: 0
统计数据: 通道2
帧率R: 0
帧率T: 0
校验错误: 0
序号 系统时间 时间标识 CAN通道 传输方向 ID号 帧类型 帧格式 长度 数据

2.4 Receiving messages



Checking the "Turn on CAN reception" option starts the reception of data on the specific channel of the enabled USBCAN adapter.

Channel 1 and Channel 2 reception is turned on by changing the relevant channel check status under Receive Enable in the Parameter Setting menu.

2.5 CAN Parameters

2.5.1 CAN parameter setting (internal function)

If the USBCAN adapter has been turned on, tap "Parameter Setting" "CAN Parameter Setting" menu item to enter the USBCAN adapter parameter setting interface.

Each channel of USBCAN adapter has a set of independent parameters, which can be set and operated independently. In the parameter setting interface, you can change the parameters of the target channel to be modified by selecting "CAN channel":

Note: This function is an internal function, when users configure the parameters, they can directly configure the filter rate and other parameters in the menu "Device

Operation"->"Startup Device" interface.

2.5.1.1 Baud rate setting

The baud rate parameter drop-down list provides standard baud rate settings from 10k to 1000k, and provides unconventional baud rate settings, select "self define" in the list, and set the baud rate register values.

Among them, the baud rate calculation formula is:

CAN baud rate = 16000000/(synchronization segment + propagation time segment + phase buffer segment 1 + phase buffer segment 2)/pre-divided frequency;

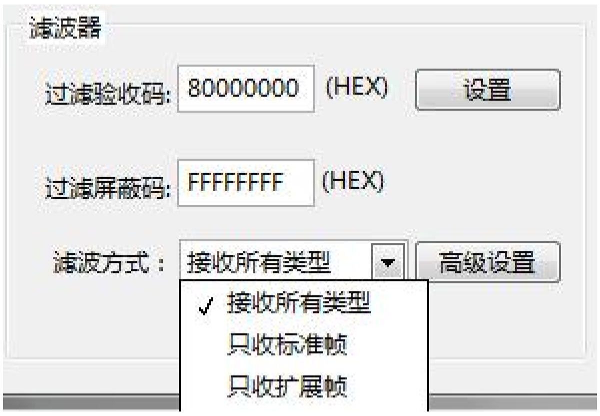
2.5.1.2 Operating modes

Normal operating mode, monitor-only listening mode and self-test mode (loopback mode) can be selected:

2.5.1.3 Filter Settings

When the module receives a message, it compares the message identifier with the corresponding bit in the filter. If the identifier matches the user-configured filter, the message is stored in the corresponding receive cache queue of the CAN controller.

The receive mask can be used to ignore the select bits of the identifier during reception. These bits will not be compared to the bits in the filter when the message is received. For example, if the user wishes to receive all messages with identifiers 0, 1, 2 and 3, the user needs to mask out the lower 2 bits of the identifier.



The USB-CAN bus adapter provides three types of filtering, which are described in the table below:

Value	Name	Instruction
3	Receive All Types	The filter filters both standard and extended frames!
4	Standard Frames Only	The filter only filters standard frames, extended frames will be filtered out directly.
7	Extended Frame Only	The filter only filters extended frames, standard frames will be filtered out directly.

Note: For detailed description of baud rate, working mode, filter settings, etc., please refer to "9.

Click the "Advanced Settings" button in the Filter Settings area to enter the Filter Advanced Settings interface, as shown in the following figure:

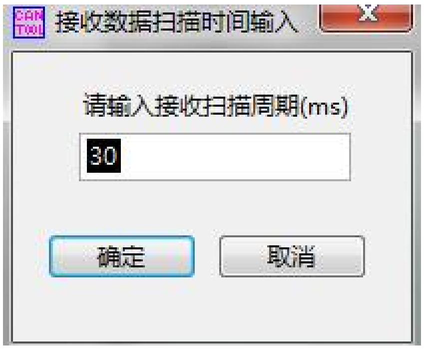
The user can make customized settings for the filter in this interface to provide convenience!

2.6 Other Settings

2.6.1 Receive Scan Time

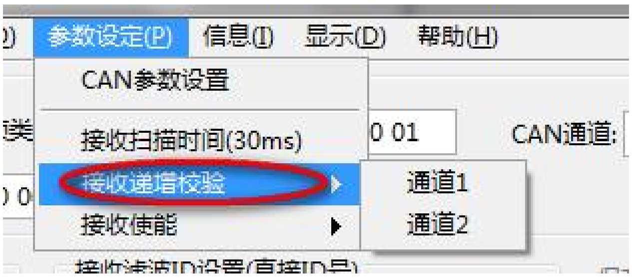
Tap "Parameter Settings" "Receive Scan Time (xx ms)" menu item to modify the minimum scan period of CAN reception, where xx denotes the currently set scan period in milliseconds.

The default (optimal) value of the receive scan period is 30ms.



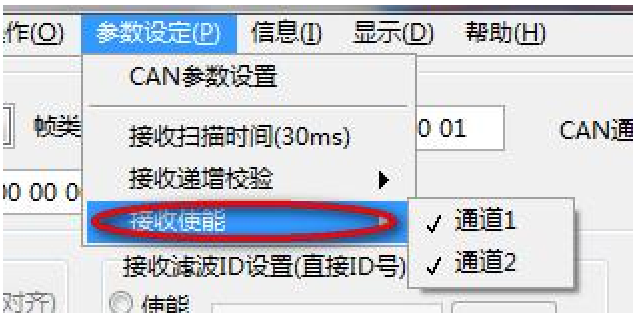
2.6.2 Incremental check

Tap the corresponding menu item in "Parameter Settings" "Receive Incremental Verification" to enable or disable the receive data incremental verification function of "Channel 1" or "Channel 2", which is used in conjunction with the "Data Increment" function when sending data. "This function is used in conjunction with the "data increment" function when sending data, and is often used to test the frame loss rate when sending and receiving CAN messages at high speed.



2.6.3 Channel Receive Enable

Tap the corresponding menu item in "Parameter Settings" "Receive Enable" to turn on or off the receive function of channel 1 or channel 2.

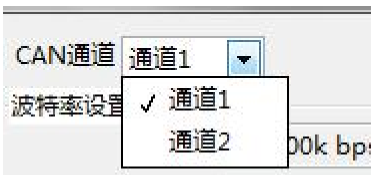


2.7 Information

2.7.1 Current Parameters

Tap the "Information" "Current Parameters" menu item to get the current parameters of the USBCAN adapter in real time.

If the device type is USB-CAN2.0, you can view the parameters of other channels by selecting "CAN channel".



2.7.2 Device Information

Tap the "Information" →"Device Information" menu item to open the device information view window.

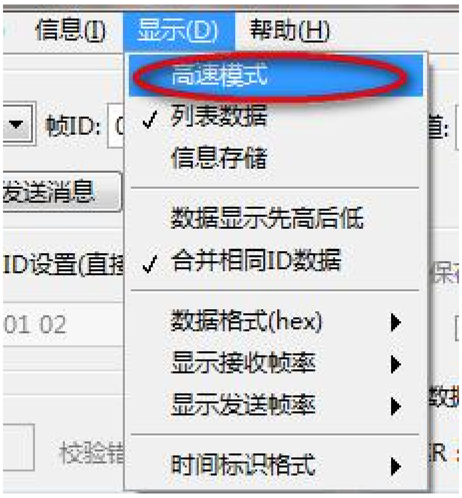
This window shows detailed device information for the currently open USBCAN adapter:

2.8 Data Display

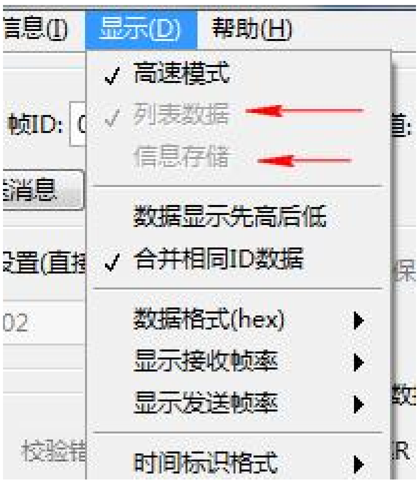
2.8.1 High-speed mode (for internal testing)

Tap "Display" "High-speed Mode" menu item to turn on or off the high-speed mode of information display, when the high-speed mode is turned on, the functions of "list data" display and "information storage" will be invalid. When the high speed mode is turned on, the two functions of "List Data" display and "Information Storage" will be invalidated, and this function is generally used for high speed transceiver test.

Note: When the frame rate of USBCAN adapter is high (e.g. 8000 frames/sec), the general list display will be relatively slow and take up computer resources, which will affect the efficiency of USBCAN adapter, and when the "High Speed Mode" is turned on, the USB-CAN Tool will only deal with the data receiving, data transmitting, data verification, and no longer deal with the data display, When "High Speed Mode" is turned on, the USB-CAN Tool will only handle data reception, data transmission, data verification, and no longer handle time-consuming operations such as data display and data storage!



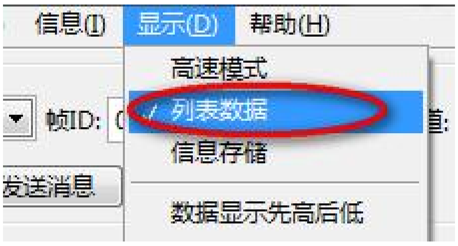
The following figure illustrates the situation when the related items are disabled after the high-speed mode is enabled:



2.8.2 List Data

Tap the "Display" "List Data" menu item to switch on or off the display of CAN messages in the data list. Displaying real-time messages in the data list will affect the processing speed of the software for CAN messages to a certain extent.

Note: This function is disabled when "High Speed Mode" is turned on!

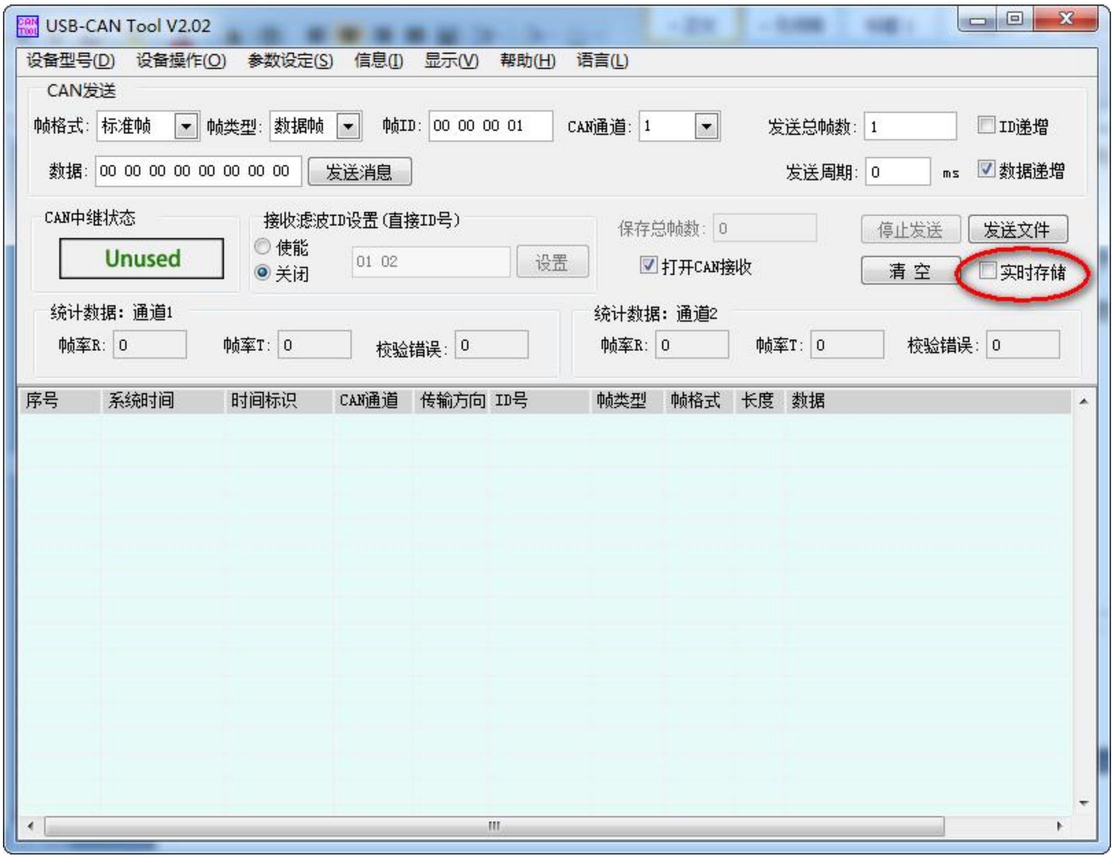


2.8.3 Real-time message storage

Tick the "Real-time storage" checkbox in the software interface to choose whether to store real-time message information into an Excel document. When the real-time message storage function is enabled, the "File Save Path Selection" dialogue box will pop up, asking the user to select the directory of the file to be stored and the name of the file.

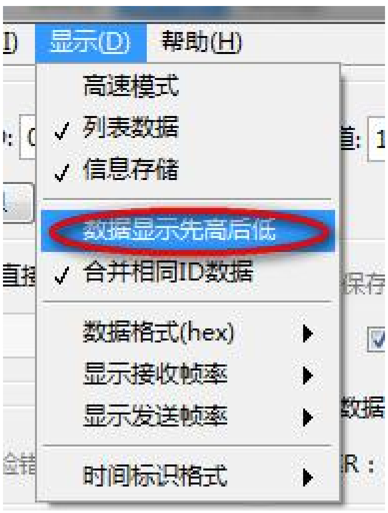
When the real-time information storage function is enabled, "_xxx" will be appended to the name of the file selected by the user, where xxx is the serial number of the sub-file, 000 is the first sub-file, 001 is the second sub-file, and so on. Each sub-file holds 10000 frames of data.

Note: This function is disabled when "High Speed Mode" is turned on!



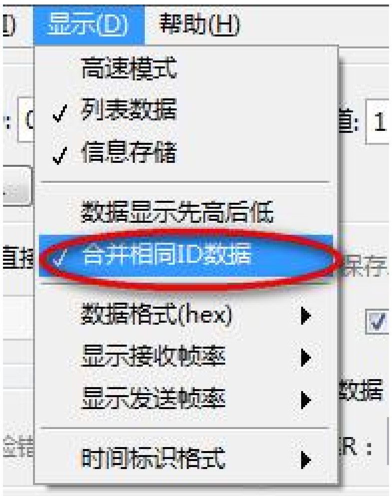
2.8.4 Byte Order

You can set the order of bytes when displaying data in the list by clicking the menu item "Display" and "Data display first high, then low".



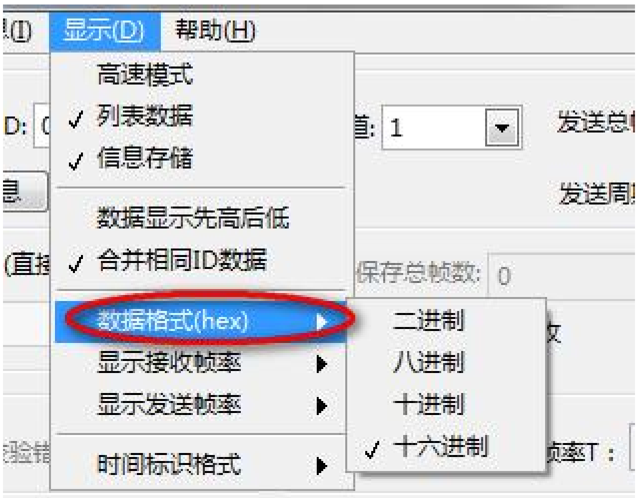
2.8.5 Merge Same ID Data

Tap "Display" → "Merge Same ID Data" menu item to turn on or off the function of merging same ID data, when this function is turned on, the CAN message frames with the same ID will only display the latest frame in the list.



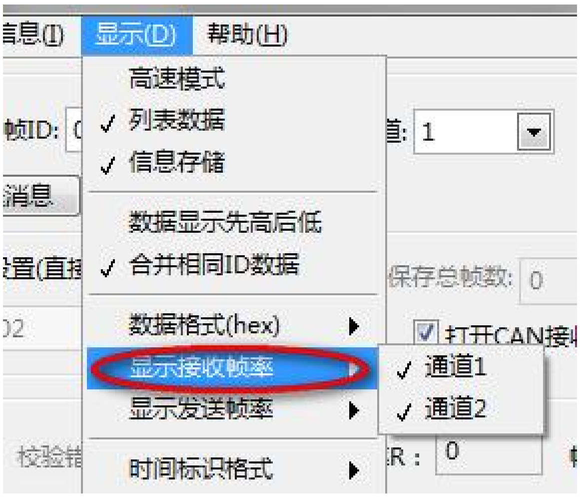
2.8.6 Data format

Tap the corresponding item in the "Display" "Data Format (xxx)" menu item to change the data display format in the current CAN message list, the available formats are: binary, octal, decimal, hexadecimal, where xxx indicates the abbreviation of the current format. The available formats are: binary, octal, decimal, hexadecimal, where xxx is the abbreviation of the current format.



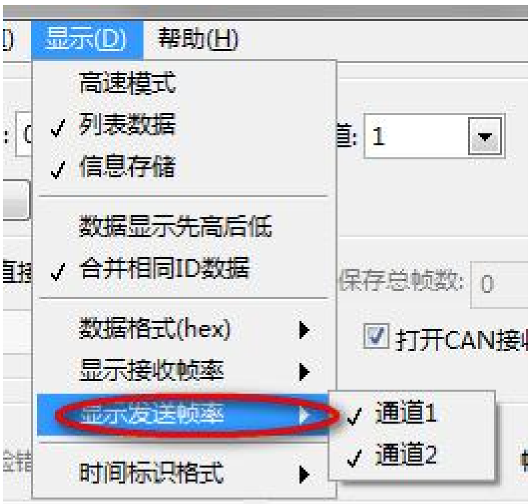
2.8.7 Receive frame rate display

Tap the channel number in the "Display" "Display Receive Frame Rate" menu item to turn on or off the receive frame rate display.



2.8.8 Transmit frame rate display

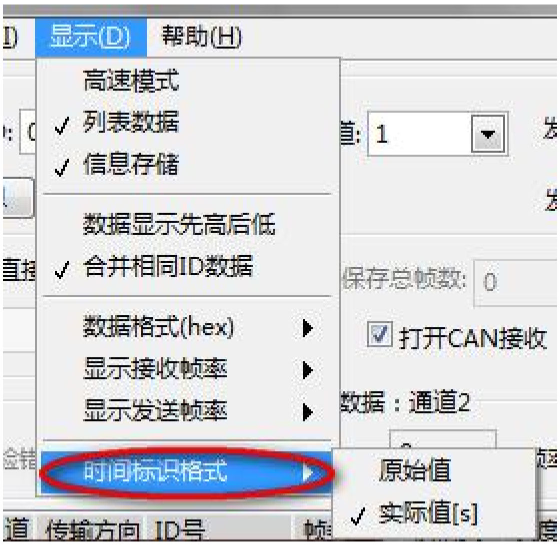
Tap the channel number in the menu item "Display" "Display transmit frame rate" to turn on or off the display of transmit frame rate.



2.8.9 Time Tag Format

A time mark is a time stamp placed on each frame of data received that records the exact time the USBCAN device received the frame, starting at the initialized USBCAN device time. It is generally used for precise plotting of the time axis.

Click the "Display" "Time Label Format" menu item, and select the time label format as the original value or the actual value. The unit of actual value is 1s, and the unit of original value is 0.1ms.



2.9 Data reception software filtering

In the Receive Filter ID Setting area, tap Enable to modify the Receive Filter ID value and set it to turn on the Receive Filter Setting, which compares the CAN message frame ID read from the USBCAN adapter with the set Filter ID group, and displays and processes the ones that are allowed to be received, otherwise they are discarded.

The ID values allowed to be received can be set to more than one, expressed in hexadecimal and separated by spaces, e.g., if the Receive ID is set to "18FF05C8 03 9C8", it means that the USB-CAN Tool will only process the CAN message frames with ID=0x18FF05C8, 0x03, and 0x9C8, and will display and process the frames with other IDs. other IDs are ignored.

2.10 Data display list

When the "List Data" menu is checked and not in "High Speed Mode", the data display list will show the contents of the CAN messages being received and sent in real time.

序号	系统时间	时间标识	CAN通道	传输方向	ID号	帧类型	帧格式	长度	数据
00000	21:22:58.767	无	ch1	发送	0001	数据帧	标准帧	8	x| 00 00 00 00 00 00 00 00
00001	21:22:58.765	2348.9228	ch2	接收	0001	数据帧	标准帧	8	x| 00 00 00 00 00 00 00 00

2.10.1 Serial number

The position of the CAN message in the message cache, starting from 0. Up to 100,000 frames of data are stored in the data cache.

2.10.2 System time

The system time at which the CAN message was sent or received, which is obtained from the computer.

2.10.3 Time identification

Hardware time when CAN message is received, unit: second, the time starts from the startup of USB-CAN device, the accuracy is 0.1 millisecond. (Tap "Display" "Time mark format" menu item to select the time mark format as original value or actual value. (The unit of the actual value is 1s, and the unit of the original value is 0.1ms.)

2.10.4 CAN channel

The CAN device channel number used when CAN messages are sent or received, ch1 means channel 1 and ch2 means channel 2.

2.10.5 Transmission direction

Marks whether the CAN message is sent or received.

2.10.6 ID number

The ID value of the CAN message. For an introduction to the ID, please refer to: "Annex 1: Detailed description of ID alignment" description document.

2.10.7 Frame type

Marks whether the CAN message is a data frame or a remote frame.

2.10.8 Frame format

Indicates whether the CAN message is a standard or extended frame.

2.10.9 Length

The data length of the CAN message in bytes.

2.10.10 Data

Data of the CAN message. Refer to section 2.8.4 Byte order and section 2.8.6 Data format for how the data is displayed.

2.11 Convenient Auxiliary Functions

2.11.1 Automatic saving of configuration information

After the software runs, the interface state at the end of the last time is automatically loaded, including menu item settings and interface control values.